

I am a creative, versatile, and driven engineer capable of hybridizing robotics, optimization and mechanical design. I desire to improve the built environment by advancing human machine collaboration and robot manipulation.

Education

BASc, Engineering Physics (ENPH), 89% GPA
The University of British Columbia, Vancouver, B.C.

September 2014 – December 2019

Engineering Experience



Genesis Robotics and Motion Technologies
Concept Design Engineer

January 2021 – August 2022

- Co-developed a series of joints (knee, ankle, hip) for a large legged robot application, utilizing a torque dense motor and a compact differential planetary gear topology to achieve 400 Nm continuous torque.
- Increased gearbox simulation process throughput by 1000x by applying an innovative big data approach to performance prediction, writing a custom 'gear prediction pipeline' (GPP) python library.
- Reduced power loss of prototype from 40% to 15%, solving efficiency problems by implementing an evolutionary optimization algorithm to select gear geometry for low wear operation.
- Analyzed insights from GPP datasets and generated intuitive visualizations to refine business case and market scope of technology by invalidating initial thermal and cost product requirements.

Gear Development Engineer

May – August, 2019, March – December, 2020

- Invented a lightweight, cheap compound planetary reducer with no carrier, filed a provisional patent, and validated packaging requirements by designing and assembling prototypes.
- Developed a test bench for characterizing crucial attributes (stiffness/backlash) of robotic actuator prototypes, automated data processing to quickly generate actionable reports from raw data.



Tesla, Inc.
Service Engineering Intern

May – December, 2018

- Quantified accumulated wear on vehicle components by building regression based prognostic models from fleet wide data analysis of use cases on over 100 000 cars.
- Summarized fleet wide usage of vehicle sensors in collaboration with design team seeking to eliminate redundant components and reduce cost of product.
- Diagnosed critical, high severity vehicle failures in less than 24 hours, concisely contributing findings to a daily email report used by CEO to drive critical quality control improvements on production line.
- Repaired end of line vehicles in company wide effort to produce 5000 Model 3's produced per week, and supervised teams performing 800 vehicle inspections pre-delivery to achieve profitable quarters.



A&K Robotics
Mechanical Design Intern

May 2017– April 2018

- Increased lifetime of a janitorial robot drive system by 40x, co-developing a novel chain based power transmission for cost effective, robust field operation.
- Oversaw prototyping, fabrication and maintenance of 2 prototypes during A&K's first fully autonomous cleaning field trials at a major transit centre and international airport.

Design Coursework

ENPH 479 (for TRIUMF Ultra Cold Neutrons Group)

June 2019 – February 2020

- Earned an APSC innovation award for effectively demonstrating functionality of project to 50 fellow students, judges, and sponsor.
- Lead a team of 3 to design a non-ferrous, remotely actuated, 3 axis gantry system with a 1mm positioning accuracy for an automated magnetic field metrology device.

ENPH 459 (for NZ Technologies, Inc.)

September – April 2018

- Co-developed a low cost, lightweight 4 DoF robot arm (SCARA) for positioning a gesture recognition device in operating rooms, utilizing waterjet cutting, 3D printing, and machining to construct stiff joints.
- Reduced end effector deflection by 4x of the previous prototype to achieve <1mm compliance.

Volunteer Experience

UCM at UBC

September 2017 – December 2019

- Planned and facilitated weekly meetings for 8 students, providing a space to explore faith based questions in the context of a secular undergraduate education.
- Directed games nights for over 60 members, aimed at fostering relationships between students by providing an escape from academic stress.

ENPH Mentorship Program

September – December 2017

- Collaborated with co-mentor in coordinating weekly meetups with 8 incoming students, creating a safe space to discuss stress management, and providing mentorship through job application/resume reviews.

Quantum Degenerate Gases Laboratory

October 2015 – September 2017

- Protected a \$40,000 laser system from damage by designing an interlock monitoring system, automating safe shut down of amplifiers upon loss of an input power signal.
- Automated an ultra-cold atom beam shutter, utilizing stepper motor and custom 3D printed, gear based power transmission, delivering 5 years of error free operation.

Skills

Big Data Analytics	Python (Matplotlib, Plotly, Numpy, Scipy, Pandas, Sklearn), Spark, AWS EC2/S3, SQL
Machine Learning	Regression, Evolutionary/Gradient Based Optimization, Prognostic Failure Modelling
Mechanical Design	Solidworks, Onshape, Fusion 360, ASME Y14.5-2018 GD&T, Tolerance Stack Up
Prototyping	Waterjet/Laser Cutter, 3D Printers, Manual Machines (Mill/Lathe)
Mechanical Analysis	Gearbox Dynamics/Efficiency Modelling, Classical Mechanics, Mechanics of Materials

Awards

APSC Design and Innovation Day Industry Award for ENPH 479	2020
Dean's Honour List	2014 - 2019
Trek Excellence Scholarship	2016, 2018
ENPH 253 Robotics Competition, 1 st Place	2016
McElhanney Senior Design Competition, 2 nd Place	2016
EGBC Post-Secondary Entrance Scholarship	2014
Top IB Physics HL Student Award	2014